Women's Transport Intelligence
Real-Time Multi-Agent System

Complete Brief (0→Delivery)

PROJECT OVERVIEW

What we're building:	A real-time, learning, multi-agent intelligence system (not a chatbot)
Key difference:	Learns from behavior, predicts needs, coordinates 7 specialized agents
Like:	Grok/Muse but proactive, learning, multi-agent (not just conversational)
SDG 5 alignment:	Removes barriers restricting women's transport freedom
Timeline:	72 hours (4 people)
Architecture:	7 AI agents + n8n orchestration + real-time learning + multi-interface

THE 7 AGENTS

1. Route Intelligence Agent: Finds optimal routes considering all constraints

2. Safety Inference Agent: Real-time safety scoring for routes, stops, times

3. Constraint Satisfaction Agent: Understands & handles real women's constraints

4. Accessibility Agent: Ensures system works for all women (device, language, literacy)

5. Learning & Memory Agent: Builds persistent profiles, learns from behavior

6. Prediction Agent: Anticipates needs before women ask

7. Orchestrator Agent: Coordinates all agents, synthesizes responses

HOW IT WORKS: REAL EXAMPLE

User: Priya, 28, feature phone, travels with 5yo daughter, works 9-5, scared of waiting alone

Day 1	SMS: 'How to get to Nungambakkam with my daughter?' → System creates profile, filters routes, scores safety, sends: Rou
Day 2	SMS: 'What about 8:15? Daughter has school at 8:30.' → System recognizes tight deadline, predicts Route 5 won't work, su
Week 1	System detects flooding on Route 5, proactively sends alert: 'Flooding near Route 5 tomorrow. Try Route 3 instead.' → Priy
Week 2+	System now predicts her entire routine. Less reactive, more intelligent.

PHASE 1: RESEARCH (24 hours | Day 1)

Goal: Understand barriers. Build agent system for real problems, not generic ones.

• **Step 1 (2h):** 20-question survey covering demographics, constraints, proactive needs, digital access, systemic barriers

• **Step 2 (4h):** Distribute to 60-100 women (SRM groups, Chennai communities)

• **Step 3 (6h):** Claude clusters responses → identifies 5-7 personas, barriers, proactive needs

• **Step 4 (1h):** Decide: Does multi-agent approach solve this?

PHASE 2: MULTI-AGENT DESIGN (28-32 hours | Day 1.5 - Day 2.5)

Goal: Prompt Claude/Gemini to generate entire agent system.

• **Step 1 (4h):** For each agent, prompt Claude → get specs, logic, learning mechanism

• **Step 2 (3h):** System architecture: how agents coordinate, data flows, real-time sync

• **Step 3 (2h):** Data schemas: user profiles, real-time state, learning data, routes, incidents

• **Step 4 (4h):** n8n workflows: User handler, Orchestrator, Real-time prediction, Learning update

• **Step 5 (6h):** Agent prompts for Claude/Gemini (system prompts for each agent)

• **Step 6 (6h):** Multi-interface: SMS, voice, WhatsApp, web adapters + formatters

• **Step 7 (2h):** Real-time data sources (mock + API stubs)

PHASE 3: BUILD & DEPLOY (18-24 hours | Day 2.5 - Day 3)

• **Step 1 (3h):** Infrastructure: n8n, storage (Firebase/JSON), Claude/Gemini setup, Sarvam

• **Step 2 (8h):** Implement 7 agents in sequence (Route → Constraint → Safety → Accessibility → Learning → Prediction → Orchestrator)

• **Step 3 (4h):** Multi-interface: SMS gateway, voice, WhatsApp bot, web dashboard

• **Step 4 (3h):** Real-world testing with 10-15 women (verify learning, proactivity)

• **Step 5 (3h):** Polish, demo video (3-4 mins: learning + proactivity)

PHASE 4: PRESENTATION (8-10 hours | Day 3)

Narrative arc:

1. **Research:** 'We asked 75 women. Here's what they actually need...'

2. **Gap:** 'Existing bus apps are dumb—they don't learn, don't anticipate, don't adapt.'

3. **Solution:** '7 agents working together. One understands constraints. One learns. One predicts.'

4. **Demo:** Show Priya's journey (learning + proactivity) across SMS → next day → week.

5. **Impact:** 'SDG 5 = removing barriers. This system does that by understanding women's real lives.'

15 SLIDE DECK

• 1. Title: SafePath Intelligence | Real-Time Multi-Agent System

• 2. Problem: Research findings + personas

• 3. Gap: Why chatbots fail (dumb, passive, generic)

• 4. Solution overview: Multi-agent intelligence

- 5-7. Agent showcase (Route, Safety, Accessibility, Learning, Prediction)
- 8. Architecture diagram (agents + coordination)
- 9. Real example: Priya's journey
- 10. Demo 1: SMS interaction + learning
- 11. Demo 2: Proactive alert
- 12. Demo 3: Web dashboard + profile
- 13. Impact: How system enables freedom
- 14. Tech stack (Claude + Gemini + n8n, brief)
- 15. Vision: 'Every woman has personal transport intelligence'

SUBMISSION DELIVERABLES

✓ Research data + 5-7 personas (PDF)

✓ Multi-agent architecture diagram (visual)

✓ System design document (PDF)

✓ Working prototype (live: SMS, voice, WhatsApp, web)

✓ Demo video (3-4 mins: learning + proactivity + real-time)

✓ Slide deck (15 slides)

✓ 2-page summary (problem → solution → impact)

TEAM (4 PEOPLE)

Person	Phase 1	Phase 2	Phase 3	Phase 4
1	Survey + analysis	Claude: agent specs	Deploy SMS/voice	Demo video
2	Data cleanup	Claude: architecture	Deploy WhatsApp	Slides
3	Mock data	Claude: n8n workflows	Deploy web	Narrative
4	Personas	Claude: system prompts	Real testing	Summary

SUCCESS CRITERIA (72 hours)

✓ 7 AI agents designed + working

✓ Multi-interface system (SMS, voice, WhatsApp, web)

✓ Learning system that improves with interactions

✓ Proactive alerting working

✓ Real-world test: women using it, system learning

✓ Clear narrative: research → design → demo → impact

✓ Judges see: Not just a feature. A system.

DECISION POINTS (Decide NOW)

1. All 7 agents vs 5 core agents?
2. Full persistent profiles vs session learning?
3. Mock data only vs 1-2 real API integrations?
4. Dazzle judges (all features) vs solid execution?

RESOURCES

✓ Claude Pro (you have)

✓ Gemini API (free tier)

✓ Sarvam AI (check cost)

✓ n8n (free tier)

✓ Firebase/Supabase (free)

✓ Vercel/Netlify (free)

Budget: ■0-1000

START WITH RESEARCH. EVERYTHING ELSE FOLLOWS.

'Every woman should have a personal transport intelligence that learns who she is, understands what she needs, and adapts in real-time.'